

EC2020: Chapter 3: Simple Regression Model

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Definition of a Simple Regression Model

The simple regression model studies how y varies with changes in x .

For a population, we can express the relationship as:

$$y = \beta_0 + \beta_1 x + u$$

- The dependent variable is y
- The explanatory variable is x
- The error term is u
- The parameter β_0 is the intercept, and β_1 is the slope

Definition of a Simple Regression Model

You can think of x as the **observed** variable and u as the **unobserved** variable.

For example, a survey might ask "*how many years of formal education have you had?*". This data is collected and therefore we can observe it.

But education is not the only factor that influences earnings. It can also be your profession, industry, parent's education, marital status, etc.

If we don't have information on these variables, then they represent all the unobserved factors.

Definition of a Simple Regression Model

Throughout the module, we will learn about strategies of mitigating this issue.

For now, let's assume that we have **zero conditional mean**:

$$E[u|x] = E[u] = 0$$

This means that the error term has the same expectations given any value of the explanatory variable.

For example, if u includes a person's ability, then we are assuming that $E[abil | educ = 12] = E[abil | educ = 16]$.

Definition of a Simple Regression Model

If we apply the zero conditional mean assumption on our simple linear model:

$$\begin{aligned} E(y|x) &= E\left[\beta_0 + \beta_1 x + u\right] \\ &= E\left[\beta_0 + \beta_1 x + u|x\right] \\ &= \beta_0 + \beta_1 x + E[u|x] \\ &= \beta_0 + \beta_1 x \end{aligned}$$

Which is the population regression line.

Deriving the OLS Estimates

We do not observe the population, which means that the parameters β_0 and β_1 are unknown.

However, if we have a sample of the population, we can estimate the parameters as $\hat{\beta}_0$ and $\hat{\beta}_1$.

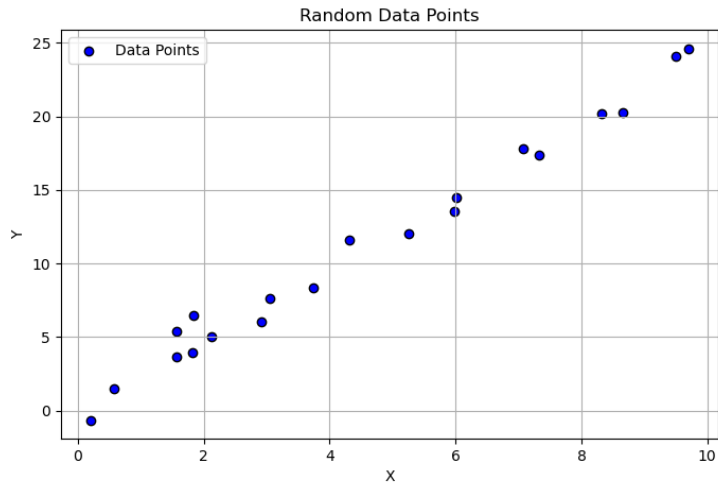
The estimator we will use is called Ordinary Least Squares.

Deriving the OLS Estimates

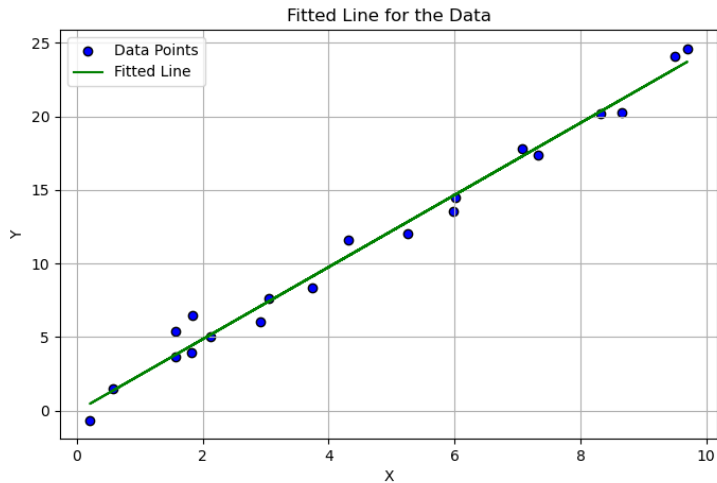
Ordinary Least Squares creates **estimators** of the parameters by **minimising the residual sum of squares**:

$$\min \sum_{i=1}^N \hat{u}_i^2$$

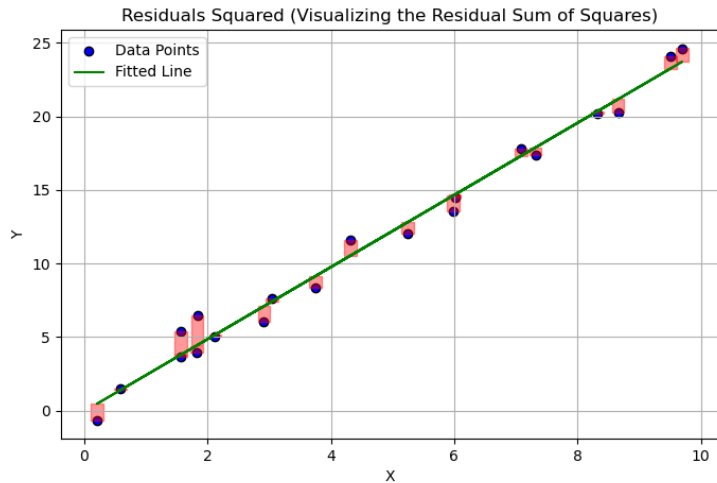
Deriving the OLS Estimates



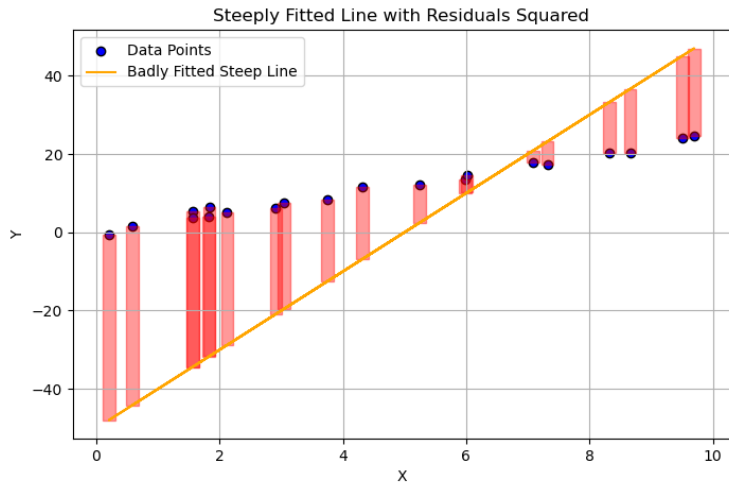
Deriving the OLS Estimates



Deriving the OLS Estimates



Deriving the OLS Estimates



Deriving the OLS Estimate

Mathematically, we can define this difference as the residual term of each observation:

$$\hat{u}_i = y_i - \hat{y}_i$$

Which can be expanded as:

$$\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

Deriving the OLS Estimate

The objective of the optimisation process is find values b_0 and b_1 that **minimise the residual sum of squares**.

These parameters are continuously adjusted until the optimal values $\hat{\beta}_0$ and $\hat{\beta}_1$ have been found. This occurs when we take first order conditions.

$$\min_{b_0 b_1} \sum_{i=1}^N \hat{u}_i^2 = \sum_{i=1}^N (y_i - b_0 - b_1 x_i)^2$$

Deriving the OLS Estimate

Take the partial derivatives with respect to the parameters:

$$\frac{\delta \sum u_i^2}{\delta b_0} = \sum_{i=1}^N -2(y_i - b_0 - b_1 x_i)$$

$$\frac{\delta \sum u_i^2}{\delta b_1} = \sum_{i=1}^N -2x_i(y_i - b_0 - b_1 x_i)$$

Deriving the OLS Estimate

$\hat{\beta}_0$ and $\hat{\beta}_1$ will be the values of b_0 and b_1 that satisfy first order conditions:

$$\frac{\partial \sum u_i^2}{\partial \hat{\beta}_0} = 0 \quad \text{and} \quad \frac{\partial \sum u_i^2}{\partial \hat{\beta}_1} = 0$$

Therefore:

$$= \sum_{i=1}^N -2(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$= \sum_{i=1}^N -2x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Deriving the OLS Estimate

Solving for the first equation:

$$= \sum_{i=1}^N -2(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$= \sum_{i=1}^N (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$= \sum_{i=1}^N y_i - \sum_{i=1}^N \hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^N x_i = 0$$

$$= n\bar{y} - n\hat{\beta}_0 - n\hat{\beta}_1\bar{x} = 0$$

Deriving the OLS Estimate

Making $\hat{\beta}_0$ the subject:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

We can substitute this in the second equation and solve:

$$\begin{aligned} &= \sum_{i=1}^N -2x_i(y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i) = 0 \\ &= \sum_{i=1}^N x_i(y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i) = 0 \\ &= \sum_{i=1}^N x_i y_i - \bar{y} \sum_{i=1}^N x_i + \hat{\beta}_1 \bar{x} \sum_{i=1}^N x_i - \hat{\beta}_1 \sum_{i=1}^N x_i^2 = 0 \end{aligned}$$

Deriving the OLS Estimate

$$= \sum_{i=1}^N x_i y_i - n \bar{y} \bar{x} + n \hat{\beta}_1 \bar{x}^2 - \hat{\beta}_1 \sum_{i=1}^N x_i^2 = 0$$

$$= \sum_{i=1}^N x_i y_i - n \bar{y} \bar{x} = \hat{\beta}_1 \sum_{i=1}^N x_i^2 - n \hat{\beta}_1 \bar{x}^2$$

$$= \sum_{i=1}^N x_i y_i - n \bar{x} \bar{y} = \hat{\beta}_1 \left(\sum_{i=1}^N x_i^2 - n \bar{x}^2 \right)$$

Deriving the OLS Estimate

Re-arranging, this gives us:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N x_i y_i - n \bar{y} \bar{x}}{\sum_{i=1}^N x_i^2 - n \bar{x}^2} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

Deriving the OLS Estimates

Proof for the numerator:

$$\begin{aligned} &= \sum_{i=1}^N x_i y_i - n \bar{x} \bar{y} = \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) \\ &= \sum_{i=1}^N x_i y_i - \sum_{i=1}^N x_i \bar{y} - \sum_{i=1}^N \bar{x} y_i + \sum_{i=1}^N \bar{x} \bar{y} \\ &= \sum_{i=1}^N x_i y_i - n \bar{x} \bar{y} - n \bar{x} \bar{y} + n \bar{x} \bar{y} \\ &= \sum_{i=1}^N x_i y_i - n \bar{x} \bar{y} \end{aligned}$$

Deriving the OLS Estimates

Proof the denominator:

$$\begin{aligned} &= \sum_{i=1}^N x_i^2 - n\bar{x}^2 = \sum_{i=1}^N (x_i - \bar{x})^2 \\ &= \sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x}) \\ &= \sum_{i=1}^N x_i^2 - 2\bar{x} \sum_{i=1}^N x_i + \sum_{i=1}^N \bar{x}^2 \\ &= \sum_{i=1}^N x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\ &= \sum_{i=1}^N x_i^2 - n\bar{x}^2 \end{aligned}$$

Deriving the OLS Estimate

Going back, this just shows that

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} \\ &= \frac{\widehat{cov}(x, y)}{\widehat{var}(x)}\end{aligned}$$

Deriving the OLS Estimates

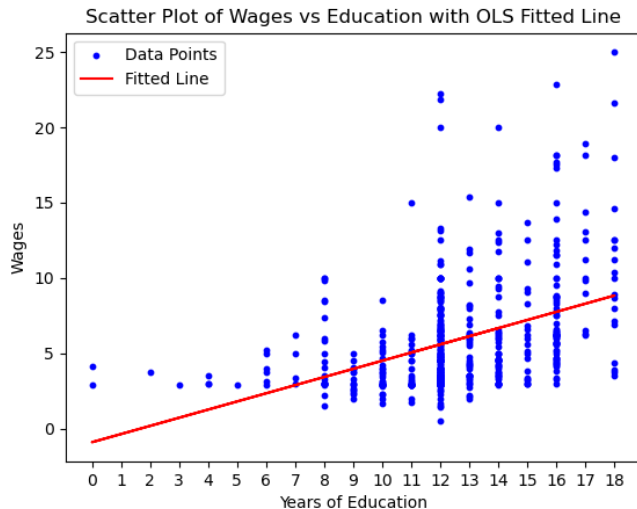
Using the WAGE1 dataset, we can compute the OLS estimates:

```
. reg wage educ
```

Source	SS	df	MS	Number of obs	=	526
Model	1179.73204	1	1179.73204	F(1, 524)	=	103.36
Residual	5980.68225	524	11.4135158	Prob > F	=	0.0000
				R-squared	=	0.1648
				Adj R-squared	=	0.1632
Total	7160.41429	525	13.6388844	Root MSE	=	3.3784

wage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.5413593	.053248	10.17	0.000	.4367534	.6459651
_cons	-.9048516	.6849678	-1.32	0.187	-2.250472	.4407687

Deriving the OLS Estimates



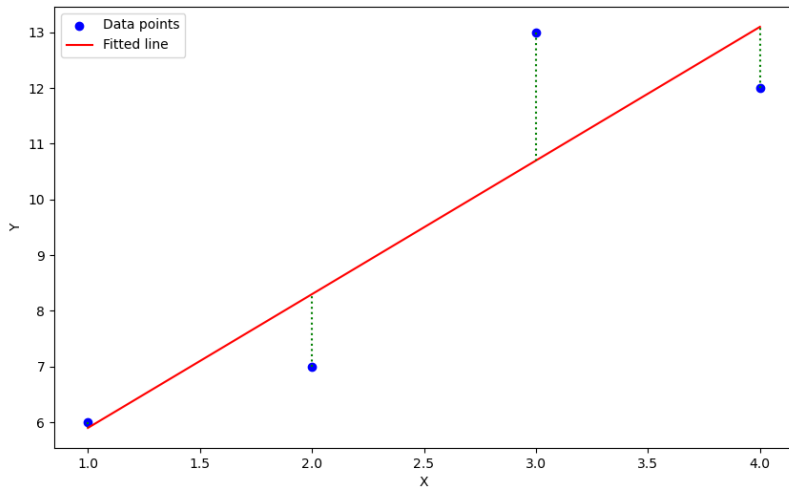
Deriving the OLS Estimates

The **red line** is known as the **fitted values**, $\hat{y}$. That is our estimate of what the population regression line looks like.

To be clear, **we do see the actual values** of y , they are the blue dots in the figure.

The difference between the true value y and the fitted value $\hat{y}$ is known as the **residuals**.

Deriving the OLS Estimates



Properties of OLS

Now that we have derived $\hat{\beta}_0$ and $\hat{\beta}_1$, we can highlight some important properties:

Property 1: OLS residuals always add up to zero

$$\sum_{i=1}^N \hat{u}_i = 0$$

Properties of OLS

Proof:

$$\hat{u}_i = y_i - \hat{y}_i$$

$$\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

$$\sum_{i=1}^N \hat{u}_i = \sum_{i=1}^N y_i - \sum_{i=1}^N \hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^N x_i$$

$$n\hat{\bar{u}} = n\bar{y} - n\hat{\beta}_0 - n\hat{\beta}_1\bar{x}$$

Properties of OLS

We can cancel out the n . Moreover, since $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$:

$$\hat{u} = \bar{y} - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 \bar{x}$$

$$\hat{u} = 0$$

Note that $\hat{u}$ is the average value of the residual. It therefore follows that:

$$\sum_{i=1}^N \hat{u}_i = n\hat{u} = 0$$

Properties of OLS

Property 2: The sample covariance between x_i and $\hat{u}_i$ is always zero

$$\widehat{cov}(x, \hat{u}) = 0$$

Intuitively, we should expect this, since the residuals are the parts of y that is not explained by x .

Proof:

First, if you remember the second restriction $E(xu) = 0$, we know the sample moment is:

$$\sum_{i=1}^N x_i \hat{u}_i = \sum_{i=1}^N x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Properties of OLS

Using this, we can show that the **numerator** of the sample correlation between x_i and $\hat{u}_i$:

$$\begin{aligned}\sum_{i=1}^N (x_i - \bar{x})(\hat{u}_i - \hat{\bar{u}}) &= \sum_{i=1}^N (x_i - \bar{x})\hat{u}_i \quad (\text{Given that } \hat{\bar{u}} = 0) \\ &= \sum_{i=1}^N x_i \hat{u}_i - \sum_{i=1}^N \bar{x} \hat{u}_i \\ &= 0 - \bar{x} \sum_{i=1}^N \hat{u}_i \\ &= 0 - 0 \quad (\text{Given that } \sum_{i=1}^N \hat{u}_i = 0) \\ &= 0\end{aligned}$$

Properties of OLS

Property 3: The sample average of y_i and $\hat{y}_i$ is the same

$$\bar{y} = \bar{\hat{y}}$$

Proof:

$$\begin{aligned}\hat{u}_i &= y_i - \hat{y}_i \\ \sum_{i=1}^N \hat{u}_i &= \sum_{i=1}^N y_i - \sum_{i=1}^N \hat{y}_i \\ n\bar{\hat{u}} &= n\bar{y} - n\bar{\hat{y}} \\ \bar{\hat{u}} &= \bar{y} - \bar{\hat{y}}\end{aligned}$$

But we know that $\bar{\hat{u}} = 0$, so that must mean $\bar{\hat{y}} = \bar{y}$

Properties of OLS

Property 4: The sample covariance of fitted values and residuals is zero

$$\widehat{cov}(\hat{y}_i, \hat{u}_i) = 0$$

Proof:

$$\begin{aligned}\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})(\hat{u}_i - \bar{\hat{u}}) &= \sum \left([\hat{\beta}_0 + \hat{\beta}_1 x_i] - [\hat{\beta}_0 + \hat{\beta}_1 \bar{x}] \right) \hat{u}_i \\ &= \hat{\beta}_1 \sum (x_i - \bar{x}) \hat{u}_i \\ &= 0\end{aligned}$$

Properties of OLS

Property 5: Point $(\bar{x}, \bar{y})$ is on the OLS regression line

$$\bar{y} = \hat{\beta}_0 + \beta_1 \bar{x}$$

Proof:

Lets look at the equation of the fitted line:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

If we set $x = \bar{x}$, then:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

Properties of OLS

But we know what $\hat{\beta}_0$ is:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Therefore:

$$\hat{y} = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 \bar{x}$$

$$\hat{y} = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 \bar{x}$$

$$\hat{y} = \bar{y} \quad \text{when } x = \bar{x}$$

Goodness of Fit

Before we start with the interpretation of the parameters, we need to have a measure how well our model is.

One way of doing this is by computing the **goodness of fit**.

Let's re-arrange the residual and express the actual values of y as:

$$y_i = \hat{y}_i + \hat{u}_i$$

Goodness of Fit

Let us define the variance of each component as:

$$SST = \sum_{i=1}^N (y_i - \bar{y})^2$$

$$SSE = \sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2$$

$$SSR = \sum_{i=1}^N (\hat{u}_i - \bar{\hat{u}})^2$$

Which is the **T**otal, **E**xplained, and **R**esidual sum of squares.

Goodness of Fit

We need to show that:

$$SST = SSE + SSR$$

Lets use the OLS properties to decompose the SST:

$$\sum_{i=1}^N (y_i - \bar{y})^2$$

Goodness of Fit

$$\begin{aligned}\sum_{i=1}^N (y_i - \bar{y})^2 &= \sum_{i=1}^N [(\hat{y}_i + \hat{u}_i) - (\bar{\hat{y}} + \bar{\hat{u}})]^2 \\&= \sum_{i=1}^N ([\hat{y}_i - \bar{y}] + \hat{u}_i)^2 \quad \text{Given that } \bar{\hat{u}} = 0 \text{ and } \bar{\hat{y}} = \bar{y} \\&= \sum_{i=1}^N ([\hat{y}_i - \bar{y}] + \hat{u}_i)([\hat{y}_i - \bar{y}] + \hat{u}_i) \\&= \sum_{i=1}^N (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^N \hat{u}_i^2 + 2 \sum_{i=1}^N (\hat{y}_i - \bar{y}) \hat{u}_i \\&= \sum_{i=1}^N (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^N \hat{u}_i^2 + 2 \sum_{i=1}^N \hat{y}_i \hat{u}_i - 2\bar{y} \sum_{i=1}^N \hat{u}_i\end{aligned}$$

Goodness of Fit

However, we know from the properties that $\sum_{i=1}^N \hat{y}_i \hat{u}_i = 0$ and that $\sum_{i=1}^N \hat{u}_i = n\bar{\hat{u}}_i = 0$. Therefore:

$$= \sum_{i=1}^N (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^N \hat{u}_i^2 + 2 \sum_{i=1}^N \hat{y}_i \hat{u}_i - 2\bar{y} \sum_{i=1}^N \hat{u}_i$$

$$= \sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^N \hat{u}_i^2$$

$$SST = SSE + SSR$$

Goodness of Fit

We can now establish the goodness of fit of our model.

The proportion of the total sum of squares that is explained by our fitted line is known as the R^2 **of the regression**:

$$R^2 = \frac{SSE}{SST} = \frac{\sum_{i=1}^N (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

R^2 can only take a value between 0 and 1.

- If 1, then the model fits y perfectly
- If 0, then there is no relationship at all
- As $R^2 \rightarrow 1$, the model gets better at explaining y

Goodness of Fit

In general, we would like a very high R^2 .

In fact, by using OLS to estimate $\hat{\beta}_0$ and $\hat{\beta}_1$, we are actually maximising R^2 :

$$R^2 = \frac{SSE}{SST} = \frac{SST - SSR}{SST} = 1 - \frac{\sum_{i=1}^N \hat{u}_i^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

Linear Transformations

Now that we know how to create estimates of our model, we can begin with interpreting the results.

Let us start with a general method of interpreting a model. Consider the below linear regression:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

In this case, we say that **a one unit increase** in x is **predicted** to have a $\hat{\beta}_1$ **unit change** in $\hat{y}$.

Linear Transformations

Consider again the regression for the WAGE1 dataset:

```
. reg wage educ
```

Source	SS	df	MS	Number of obs	=	526
Model	1179.73204	1	1179.73204	F(1, 524)	=	103.36
Residual	5980.68225	524	11.4135158	Prob > F	=	0.0000
				R-squared	=	0.1648
Total	7160.41429	525	13.6388844	Adj R-squared	=	0.1632
				Root MSE	=	3.3784

wage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.5413593	.053248	10.17	0.000	.4367534	.6459651
_cons	-.9048516	.6849678	-1.32	0.187	-2.250472	.4407687

Where *wages* are in dollars and *educ* is the number of years of formal education.

Linear Transformations

We can express our fitted model as:

$$\widehat{wage} = -0.90 + 0.54educ$$

Here, you can interpret $\hat{\beta}_0 = -0.9$ and $\hat{\beta}_1 = 0.54$.

This means a one **year** increase in education is **predicted** to increase hourly earnings by 0.54 dollars (54 cents).

Linear Transformations

The model also predicts that if a person had zero years of education, they are expected to have a negative earning of -0.9 dollars (90 cents).

But this does not make sense, as you cannot have negative earnings

The reason we literally got a negative value is because of how $\hat{\beta}_0$ was computed.

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Linear Transformations

Using the average values from the dataset:

```
. tabstat wage educ, col(stat)
```

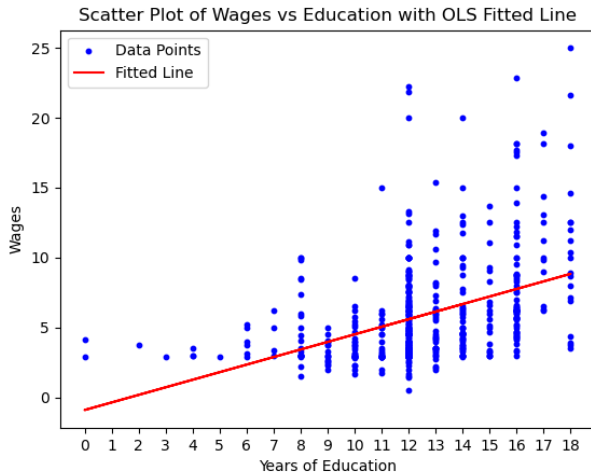
Variable	Mean
wage	5.896103
educ	12.56274

Substitute this into the expression:

$$\beta_0 = 5.89 - (0.59)12.56 = -0.9$$

Linear Transformations

The reason why we got a non-sensible intercept is because we have very few people at the lower end of our data:



Linear Transformations

Now lets consider what happens if I transform the dependent variable. Instead of producing the results using dollars, we would like to see it in cents:

$$y_i^* = \lambda y_i$$

There are 100 cents in a dollar, so:

$$\lambda = 100$$

Linear Transformations

If we recalculate $\hat{\beta}_1$ using y^*_i , then we will get:

$$\hat{\beta}_1^* = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i^* - \bar{y}^*)}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

$$\hat{\beta}_1^* = \frac{\sum_{i=1}^N (x_i - \bar{x})(\lambda y_i - \lambda \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

$$\hat{\beta}_1^* = \lambda \left(\frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} \right) = \lambda \hat{\beta}_1$$

Linear Transformations

Using this, we can also show for the intercept:

$$\hat{\beta}_0^* = \bar{y}^* - \hat{\beta}_1^* \bar{x}$$

$$= \lambda \bar{y} - \lambda \hat{\beta}_1 \bar{x}$$

$$= \lambda (\bar{y} - \hat{\beta}_1 \bar{x})$$

$$= \lambda \hat{\beta}_0$$

Linear Transformations

```
. gen new_wage = 100*wage
```

```
. reg new_wage educ
```

Source	SS	df	MS	Number of obs	=	526
Model	11797320.4	1	11797320.4	F(1, 524)	=	103.36
Residual	59806822.5	524	114135.158	Prob > F	=	0.0000
				R-squared	=	0.1648
				Adj R-squared	=	0.1632
Total	71604143	525	136388.844	Root MSE	=	337.84

new_wage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	54.13593	5.324804	10.17	0.000	43.67534	64.59651
_cons	-90.48516	68.49678	-1.32	0.187	-225.0472	44.07687

Therefore, a one additional year in education is predicted to increase a person's earnings by 54 cents (previously, 0.54 dollars).

Non-linear Transformations

Linear relationships are not general enough for typical economic applications.

One could argue that there is an *increasing* relationship between your education and your earnings.

$$wage = \beta_0 e^{\beta_1 educ}$$

If we log both sides:

$$\log(wage) = \log \beta_0 e^{\beta_1 educ}$$

$$\log(wage) = \log \beta_0 + \beta_1 educ$$

Therefore, the interpretation of β_1 is the proportional change in y per unit change of x (but we would generally multiply this by 100).

Non-linear Transformations

```
. gen log_wage = ln(wage)
```

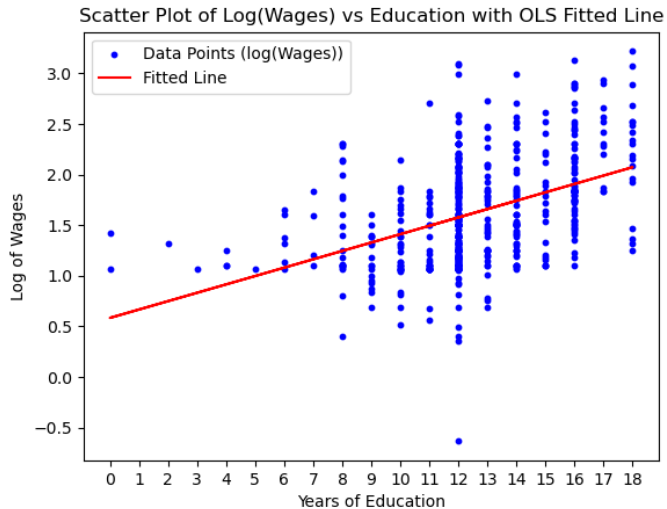
```
. reg log_wage educ
```

Source	SS	df	MS	Number of obs	=	526
Model	27.5606288	1	27.5606288	F(1, 524)	=	119.58
Residual	120.769123	524	.230475425	Prob > F	=	0.0000
				R-squared	=	0.1858
				Adj R-squared	=	0.1843
Total	148.329751	525	.28253286	Root MSE	=	.48008

log_wage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0827444	.0075667	10.94	0.000	.0678796	.0976091
_cons	.5837727	.0973358	6.00	0.000	.3925563	.7749891

We can see the *educ* is 0.08, which means that an additional year of education is predicted to increase their wages by 8% (0.083×100).

Non-linear Transformations



Non-linear Transformations

Let us consider a log-linear transformation. That is, assume that the true relationship is:

$$y = \beta_0 x_1^\beta$$

If we were to log this equation:

$$\log(y) = \log(\beta_0) + \beta_1 \log(x)$$

Here, we need to interpret the coefficients in terms of elasticity.

Non-linear Transformations

For example, let us try to model the relationship between a CEO's salary and the revenue of a company:

$$salary = \beta_0 + \beta_1 sales + u$$

Let us use the CEOSAL1 dataset to estimate the relationship:

$$\widehat{salary} = \hat{\beta}_0 + \hat{\beta}_1 sales$$

Non-linear Transformations

. reg salary sales

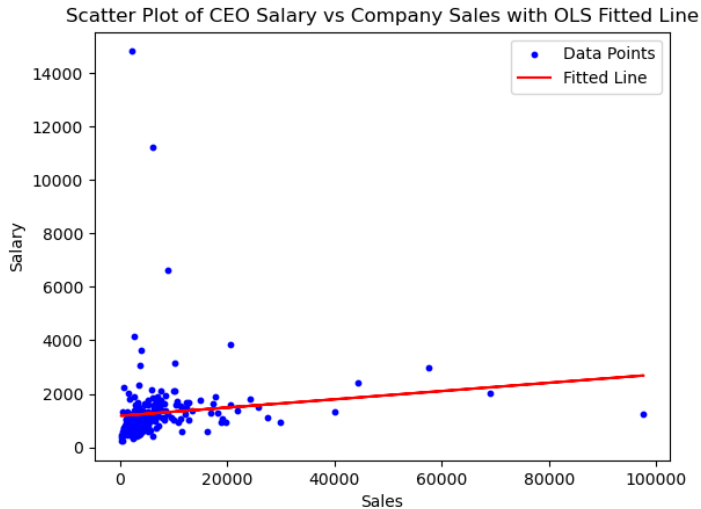
Source	SS	df	MS	Number of obs	=	209
Model	5628691.53	1	5628691.53	F(1, 207)	=	3.02
Residual	386104290	207	1865238.12	Prob > F	=	0.0838
				R-squared	=	0.0144
				Adj R-squared	=	0.0096
Total	391732982	208	1883331.64	Root MSE	=	1365.7

salary	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
sales	.0154705	.0089057	1.74	0.084	-.002087	.0330281
_cons	1174.005	112.8127	10.41	0.000	951.5958	1396.414

Where *salary* is in \$1,000 and *sales* in \$1m:

Therefore, a one thousand dollar increase in a CEO's salary is predicted to increase sales by \$15,000 ($1m * 0.015$).

Non-linear Transformations



Non-linear Transformations

It's possible that the best way to fit the model to the data is turn it into a log-log model:

$$\widehat{\log(salary)} = \hat{\beta}_0 + \hat{\beta}_1 \log(sales)$$

Non-linear Transformations

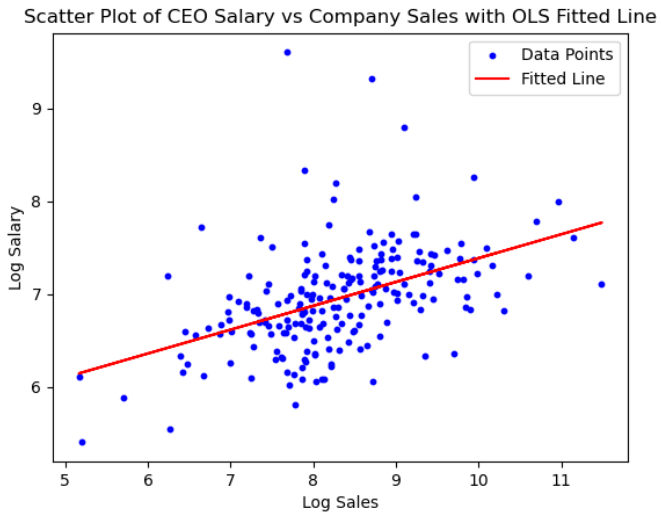
```
. reg lsalary lsales
```

Source	SS	df	MS	Number of obs	=	209
Model	14.0661688	1	14.0661688	F(1, 207)	=	55.30
Residual	52.6559944	207	.254376785	Prob > F	=	0.0000
				R-squared	=	0.2108
				Adj R-squared	=	0.2070
Total	66.7221632	208	.320779631	Root MSE	=	.50436

lsalary	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
lsales	.2566717	.0345167	7.44	0.000	.1886224	.3247209
_cons	4.821997	.2883396	16.72	0.000	4.253538	5.390455

Because this is now an elasticity model, we say that **a 1% increase in sales** is predicted to increase a CEO's salary by **0.256%**.

Non-linear Transformations



Non-linear Transformations

Level-Level Model:

$$\widehat{wage} = \hat{\beta}_0 + \hat{\beta}_1 educ$$

A one year increase in education is predicted to change *wage* by $\hat{\beta}_1$ dollars

Log-Level Model:

$$\widehat{\log(wage)} = \hat{\beta}_0 + \hat{\beta}_1 educ$$

A one year increase in education is predicted to change *wage* by $\hat{\beta}_1\%$

Non-linear Transformations

Log-Log Model:

$$\widehat{\log(\text{salary})} = \hat{\beta}_0 + \hat{\beta}_1 \log(\text{sales})$$

A 1% increase in sales is predicted to change a CEO's *salary* by $\hat{\beta}_1\%$

Level-log Model:

$$\widehat{\text{salary}} = \hat{\beta}_0 + \hat{\beta}_1 \log(\text{sales})$$

A 1% increase in sales is predicted to change a CEO's *salary* by $\frac{\hat{\beta}_1}{100}$ thousand dollars.

Assumptions

Now that we have established the estimators of $\hat{\beta}_0$ and $\hat{\beta}_1$, we need to understand the statistical properties of OLS in the population model.

Let us impose the following assumptions for our Simple Linear Regression (SLR):

Assumptions

SLR1: The model is linear in parameters

For the population model:

$$y = \beta_0 + \beta_1 + u$$

The model is 'linear in parameters' in the sense that the terms on the right hand side includes β parameters appearing separately. For example, this assumption would be violated if:

$$y = \beta_0 x^{\beta_1} + u$$

Although as we discussed in the previous section, it is possible to transform this into a linear equation using logs.

Assumptions

SLR2: Random Sampling

This states that there is a random sample of the population. We can write the population model in terms of random samples by:

$$y_i = \beta_0 + \beta_1 x_i + u_i, \quad i = 1, 2, \dots, n$$

Do note that u does not refer to residuals $\hat{u}_i$. These are separate things.

Assumptions

SLR3: Sample Variation in Explanatory Variable

This implies that x is not constant. Consider the β_1 equation:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

If x was constant, then $x = \bar{x}$, and so both numerator and denominator would be zero.

Assumptions

SLR4: Zero Conditional Mean

We impose the assumption that the disturbance term u has an expected value of zero, given any value of the explanatory variable:

$$E(u|x) = 0$$

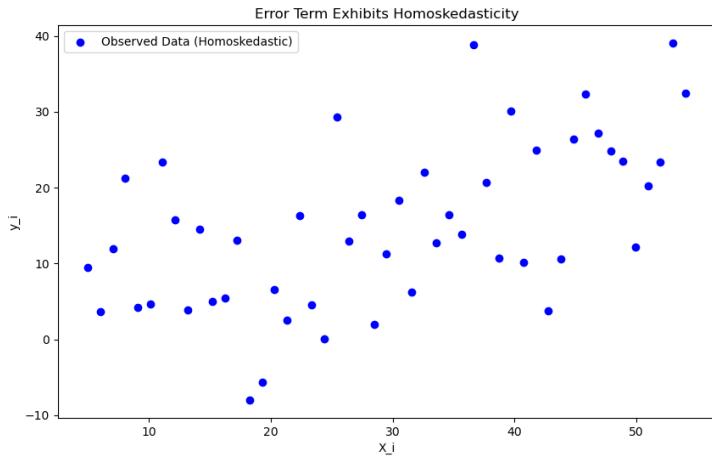
Assumptions

SLR5: Homoskedasticity

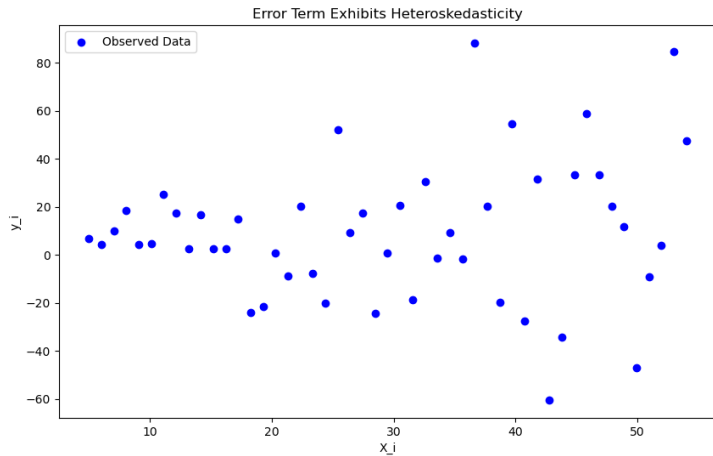
This assumes that the value of each observation drawn from the sample has a constant population variance.

This assumption helps us simplify our variance calculations. Although we will not need it to prove **unbiasdness** of OLS, it will be important when demonstrating **efficiency** of the estimator.

Assumptions



Assumptions



Assumptions

Remember that we don't actually observe σ^2 , as it is unknown.

One of the tasks of regression analysis is that we need to estimate the variance of the disturbance term.

The random components of the OLS coefficients

Our goal in this section is to prove unbiasedness of our OLS coefficients.

$$E[\hat{\beta}_0|x] = \beta_0$$

$$E[\hat{\beta}_1|x] = \beta_1$$

The random components of the OLS coefficients

Before we get there, let us change the way we write β_1 :

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

Remember that $y = \beta_0 + \beta_1 x + u$ and $\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{u}$

The random components of the OLS coefficients

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} \\&= \frac{\sum_{i=1}^N (x_i - \bar{x})(\beta_0 + \beta_1 x_i + u - \beta_0 - \beta_1 \bar{x} - \bar{u})}{\sum_{i=1}^N (x_i - \bar{x})^2} \\&= \frac{\sum_{i=1}^N (x_i - \bar{x})(\beta_1 x_i + u - \beta_1 \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2} \\&= \frac{\sum_{i=1}^N (x_i - \bar{x}) \left[\beta_1 (x_i - \bar{x}) + u \right]}{\sum_{i=1}^N (x_i - \bar{x})^2}\end{aligned}$$

The random components of the OLS coefficients

$$\begin{aligned} &= \frac{\sum_{i=1}^N \beta_1 (x_i - \bar{x})^2 + (x_i - \bar{x})u}{\sum_{i=1}^N (x_i - \bar{x})^2} \\ &= \beta_1 + \frac{\sum_{i=1}^N (x_i - \bar{x})u}{\sum_{i=1}^N (x_i - \bar{x})^2} \end{aligned}$$

According to Woolridge, $d_i = (x_i - \bar{x})$ and $SST_x = \sum_{i=1}^N (x_i - \bar{x})^2$

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^N d_i u}{SST_x}$$

The random components of the OLS coefficients

Lets prove it is unbiased:

For a population model that satisfies SLR1, we take the expected value of the slope parameter:

$$E[\hat{\beta}_1|x] = E\left[\beta_1 + \frac{\sum_{i=1}^N d_i u}{SST_x} \middle| x\right]$$

Using assumption SLR2 and SLR4, we know that $E(u|x) = 0$. This means that SST_x and d_i are constants given that we are conditioning on x . Moreover, with SLR3, $SST_x \neq 0$.

$$E[\hat{\beta}_1|X] = \beta_1 + \frac{\sum_{i=1}^N d_i E[u|x]}{SST_x} = \beta_1$$

The random components of the OLS coefficients

Therefore, we can use the law of iterated expectations to show that:

$$E(\hat{\beta}_1) = E[E(\hat{\beta}_1|x)] = \beta_1$$

The random components of the OLS coefficients

We can also show you:

$$E(\hat{\beta}_0|x) = \beta_0$$

Proof:

For a model that satisfies SLR1:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$= \beta_0 + \beta_1 \bar{x} + \bar{u} - \hat{\beta}_1 \bar{x}$$

$$= \beta_0 + \bar{u} + (\beta_1 - \hat{\beta}_1) \bar{x} \quad \text{Under SLR2 and SLR4, the average of the error is zero}$$

The random components of the OLS coefficients

Taking the expectations:

$$\begin{aligned}E(\hat{\beta}_0|x) &= E(\beta_0) + E[(\beta_1 - \hat{\beta}_1)\bar{x}|x] \\&= E(\beta_0) + (\beta_1 - E[\hat{\beta}_1|x])\bar{x} \\&= \beta_0 + (\beta_1 - \beta_1)\bar{x} \\&= \beta_0\end{aligned}$$

The random components of the OLS coefficients

Consider a cross sectional dataset of schools.

Let us estimate the relationship between math scores and the percentage of children that are eligible for free lunch:

$$math10 = \beta_0 + \beta_1 \text{Inchprg}$$

Where *math10* is the percentage of students who passed the math exam at the school, and *Inchprg* is the percentage of students who get free lunch.

The random components of the OLS coefficients

Using the MEAP93 dataset, we compute the regression:

```
. reg math10 lchprg
```

Source	SS	df	MS	Number of obs	=	408
Model	7665.26597	1	7665.26597	F(1, 406)	=	83.77
Residual	37151.9145	406	91.5071786	Prob > F	=	0.0000
				R-squared	=	0.1710
				Adj R-squared	=	0.1690
Total	44817.1805	407	110.115923	Root MSE	=	9.5659

math10	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
lchprg	-.3188643	.0348393	-9.15	0.000	-.3873523	-.2503763
_cons	32.14271	.9975824	32.22	0.000	30.18164	34.10378

So a one percentage point increase in students accessing free lunch is predicted to reduce the percentage of students passing the math exam by 0.3%. But this is probably biased as SLR4 is likely violated.

The random components of the OLS coefficients

Unbiasedness generally fails if any of our assumptions are violated:

- SLR1 requires y and x to be linearly related, with an additive disturbance. Its possible that our model is non-linear.
- SLR2 requires random sampling, but this can fail if the sample is not representative.
- SLR3 will generally hold, otherwise we cannot even compute our coefficients.
- SLR4 is required for unbiasedness, otherwise OLS will always be biased.

There is always a concern that x and u are correlated in a simple regression analysis which results in **spurious correlation**.

Variance of the OLS Estimators

Under assumptions SLR1 through SLR5:

$$\text{var}(\hat{\beta}_1|x) = \frac{\sigma^2}{SST_x}$$

Where σ^2 is a constant variance under homoskedasticity.

Variance of the OLS Estimators

Lets prove this:

$$\text{var}(\hat{\beta}_1|X) = \text{var}\left[\beta_1 + \frac{\sum_{i=1}^n d_i u_i}{SST_x} | X\right]$$

The variance of β_1 is 0 (since it is a constant):

$$\text{var}(\hat{\beta}_1|X) = \text{var}\left[\frac{\sum_{i=1}^n d_i u_i}{SST_x} | X\right]$$

Variance of the OLS Estimators

Because we are conditioning on X , we need to treat d_i and SST_x as constants. Moreover, due to Variance Property 4 (See Woolridge p.700), **the variance of the sum is the sum of the variance** due to SLR2 (u_i is iid).¹

$$\begin{aligned} \text{var}(\hat{\beta}_1|X) &= \text{var}\left[\frac{\sum_{i=1}^n d_i u_i}{SST_x} | X\right] \\ &= \frac{\sum_{i=1}^n d_i^2}{SST_x^2} \cdot \text{var}(u_i|X) \end{aligned}$$

¹If it was not true, then there would be covariance terms included in the expression $\text{cov}(u_i, u_j) \neq 0$

Variance of the OLS Estimators

Under SLR5, the variance of u is the same for all i . So it transforms into a constant σ^2

$$\begin{aligned} \text{var}(\hat{\beta}_1|X) &= \frac{\sum_{i=1}^n d_i^2}{SST_x^2} \cdot \text{var}(u_i|X) \\ &= \frac{\sum_{i=1}^n d_i^2}{SST_x^2} \cdot \sigma^2 \end{aligned}$$

Variance of the OLS Estimators

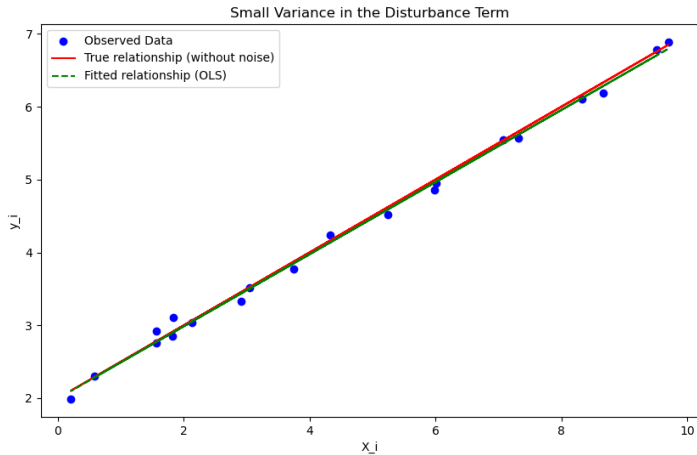
We know that $d_i = (x_i - \bar{x})$, then $d_i^2 = (x_i - \bar{x})^2$. And we also know that $SST_x = \sum_{i=1}^n (x_i - \bar{x})^2$:

$$\begin{aligned} \text{var}(\hat{\beta}_1|X) &= \frac{\sum_{i=1}^n d_i^2}{SST_x^2} \cdot \sigma^2 \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{SST_x^2} \cdot \sigma^2 \\ &= \frac{SST_x}{SST_x^2} \cdot \sigma^2 \\ &= \frac{1}{SST_x} \cdot \sigma^2 \end{aligned}$$

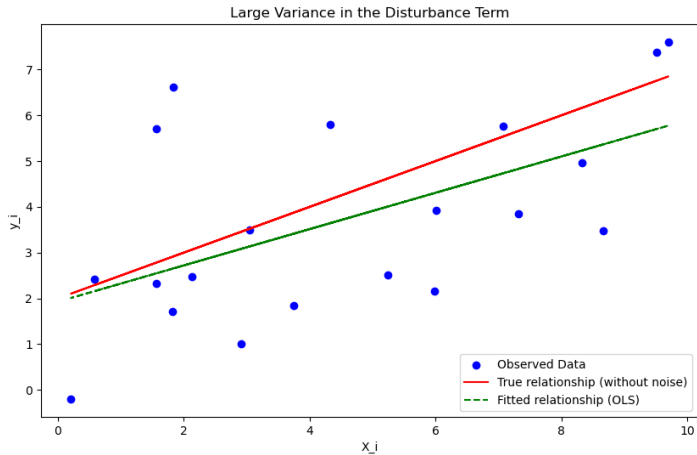
Variance of the OLS Estimators

- As $\sigma^2 \uparrow$, then the variance $\uparrow$ which makes our estimate worse.
- As $SST_x \uparrow$, then the variance $\downarrow$, which makes our estimates better.

Variance of the OLS Estimators



Variance of the OLS Estimators



Estimating the Error Variance

Now that we found the variance:

$$\text{Var}(\hat{\beta}_1|x) = \frac{\sigma^2}{SST_x}$$

We can also compute the **standard deviation** (root of the variance):

$$\text{sd}(\hat{\beta}_1) = \frac{\sigma}{\sqrt{SST_x}}$$

Estimating the Error Variance

The problem is that we do not actually see the variance term σ^2 since we do not see u . However, it is important to use the variance as we need it for hypothesis testing.

Therefore, a good alternative is to instead use the **residuals** $\hat{u}$ to create an estimate of the variance².

²Remember that there is a difference between u and $\hat{u}$. The former represents all the unobserved variables that are not accounted for in your population model whereas the latter represents the distance between your actual y and your estimated $\hat{y}$.

Estimating the Error Variance

This estimate is known as the **error variance** and it is denoted as:

$$\hat{\sigma}^2 = \frac{SSR}{n - 2}$$

Intuitively, the reason it is divided by $n - 2$ is because we are imposing two restrictions in the model³:

$$\sum_{i=1}^n \hat{u}_i = 0$$
$$\sum_{i=1}^n x_i \hat{u}_i = 0$$

³This is identical to the ideas about degrees of freedom. The appendix demonstrates why not dividing by $n - 2$ leads to a biased estimator of $\hat{\sigma}^2$.

Estimating the Error Variance

Previously, we said that the unknown **standard deviation** of the slope estimator is:

$$sd(\hat{\beta}_1) = \frac{\sigma}{\sqrt{SST_x}}$$

And since we cannot see σ , we will estimate it by using $\hat{\sigma}$:

$$se(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{SST_x}}$$

Which is known as the **standard error** of the regression. This is a value that we can compute, since $\hat{\sigma} = \sqrt{\frac{SSR}{n-2}}$

Conclusion

. reg wage educ

Source	SS	df	MS	Number of obs	=	526
Model	1179.73204 SSE	1	1179.73204	F(1, 524)	=	103.36
Residual	5980.68225 SSR	524	11.4135158	Prob > F	=	0.0000
Total	7160.41429 SST	525	13.6388844	R-squared	=	0.1648
				Adj R-squared	=	0.1632
				Root MSE	=	3.3784

wage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.5413593	.053248	10.17	0.000	.4367534	.6459651
_cons	-.9048516	.6849678	-1.32	0.187	-2.250472	.4407687
	$\hat{\beta}_1$	SE ($\hat{\beta}_1$)				
	$\hat{\beta}_0$	SE ($\hat{\beta}_0$)				

Appendix: Method of Moments

We can derive OLS using the Method of Moments, which is what is done in Woolridge.

You are expected to know both methods.

Deriving the OLS Estimates

To begin, we impose two restrictions:

$$E(u) = 0$$

$$\text{cov}(x, u) = 0$$

Since x and u are uncorrelated, then:

$$\begin{aligned}\text{cov}(x, u) &= E(xu) - E(x)E(u) \\ &= E(xu) - E(x) * 0 \\ &= E(xu)\end{aligned}$$

Deriving the OLS Estimates

Using our assumption:

$$E(u) = 0$$

$$E(xu) = 0$$

And the fact that $u = y - \beta_0 - \beta_1 x$, we substitute it into the above equations:

$$E(y - \beta_0 - \beta_1 x) = 0$$

$$E[x(y - \beta_0 - \beta_1 x)] = 0$$

Deriving the OLS Estimates

Since we don't know the population moment, we cannot use $E(\cdot)$.

However, we can estimate the expected value $E(\cdot)$ by using the sample average $\frac{1}{n} \sum_{i=1}^n$ (this is known as **the method of moments**):

$$\frac{1}{n} \sum_{i=1}^n (y - \hat{\beta}_0 - \hat{\beta}_1 x) = 0$$

$$\frac{1}{n} \sum_{i=1}^n [x(y - \hat{\beta}_0 - \hat{\beta}_1 x)] = 0$$

Deriving the OLS Estimates

Let us start with the first equation:

$$\frac{1}{n} \sum_{i=1}^N (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i = 0)$$

$$\frac{1}{n} \left(\sum_{i=1}^N y_i - \sum_{i=1}^N \hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^N x_i \right) = 0$$

$$\frac{1}{n} (n\bar{y} - n\hat{\beta}_0 - n\hat{\beta}_1 \bar{x}) = 0 \quad cont.$$

Deriving the OLS Estimates

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0$$

By setting β_0 as the subject, we get:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Which gives us the estimator of the intercept.

Deriving the OLS Estimates

Since we know the expression of $\hat{\beta}_0$, we solve for the second equation:

$$\frac{1}{n} \sum_{i=1}^n [x(y - \hat{\beta}_0 - \hat{\beta}_1 x)] = 0$$

$$\sum_{i=1}^n x_i [y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i] = 0$$

$$\sum_{i=1}^n x_i [y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i] = 0$$

$$\sum_{i=1}^n x_i (y_i - \bar{y}) - \hat{\beta}_1 \sum_{i=1}^n x_i (x_i - \bar{x}) = 0$$

$$\sum_{i=1}^n x_i (y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^n x_i (x_i - \bar{x})$$

Deriving the OLS Estimates

Which means that:

$$\sum_{i=1}^N x_i(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^N x_i(x_i - \bar{x})$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^N x_i(y_i - \bar{y})}{\sum_{i=1}^N x_i(x_i - \bar{x})}$$

Deriving the OLS Estimates

Before continuing, we can re-express both the numerator and denominator.
Starting with the numerator:

$$\begin{aligned}\sum_{i=1}^N x_i(y_i - \bar{y}) &= \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) \\ &= \sum_{i=1}^N x_i y_i - \sum_{i=1}^N x_i \bar{y} - \sum_{i=1}^N \bar{x} y_i + \sum_{i=1}^N \bar{x} \bar{y} \\ &= \sum_{i=1}^N x_i y_i - n\bar{x}\bar{y} - n\bar{x}\bar{y} + n\bar{x}\bar{y} \quad cont.\end{aligned}$$

Deriving the OLS Estimates

$$\begin{aligned} &= \sum_{i=1}^N x_i y_i - n \bar{x} \bar{y} \\ &= \sum_{i=1}^N x_i (y_i - \bar{y}) \quad \text{Note, this is also } \sum_{i=1}^N y_i (x_i - \bar{x}) \end{aligned}$$

Deriving the OLS Estimates

Now the denominator:

$$\begin{aligned}\sum_{i=1}^N x_i(x_i - \bar{x}) &= \sum_{i=1}^N (x_i - \bar{x})^2 \\ &= \sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x}) \\ &= \sum_{i=1}^N x_i^2 - 2\bar{x} \sum_{i=1}^N x_i + \sum_{i=1}^N \bar{x}^2 \quad cont.\end{aligned}$$

Deriving the OLS Estimates

$$\begin{aligned} &= \sum_{i=1}^N x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\ &= \sum_{i=1}^N x_i^2 - n\bar{x}^2 \\ &= \sum_{i=1}^N x_i(x_i - \bar{x}) \end{aligned}$$

Deriving the OLS Estimates

Going back to the original expression and substituting both our findings:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N x_i(y_i - \bar{y})}{\sum_{i=1}^N x_i(x_i - \bar{x})}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

Which is simply:

$$\hat{\beta}_1 = \frac{\text{cov}(x, y)}{\text{var}(x)}$$